Smaller explicit Diophantine forms of ZFC consistency

Compact root bounds, signed coordinates, and odd scales

An AI-assisted research contribution prepared for ProveIt

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Abstract

We combine the current ProveIt constructions with three additional reductions. The main moderate-degree result is a fully specified polynomial in twelve positive unknowns of exact degree 1097, evaluated by 116 binary arithmetic operations. Its three-equation presentation has maximum degree 1088 and costs 110 operations. This improves the reviewed twelve-variable degree minimum 1145 at 123 operations and the previous 116-operation degree 1353. The central improvement is to shift the index quotient by q and replace a large radical-separation bound by the two-operation expression $I(T+I)$. We also combine the odd-scale theorem with the existing integer lifts to obtain a quartic in 51 unrestricted integers, a sextic in 42, and an octic in 36, reducing each count by three. A separate construction gives an explicit equation in 17 unrestricted integer unknowns, compared with the 26 in the reviewed direct compiler family. It uses 20 607 operations and has exact degree $424296 \cdot 5^{33} + 53824$; this is a variable-count improvement with a very large degree cost. The coefficients for the twelve-variable and low-degree forms are the complete materialized 943-tile tuple, reproduced byte for byte. The seventeen-variable form retains the entire original package-18 T4 tuple and its finite index recipe. For each stated form, ordinary mathematics formalizable in ZFC proves that absence of solutions is equivalent to Con(ZFC). The inherited compiler and logical calibration are identified explicitly; finite audits are not represented as new proof-assistant certificates. No absolute minimality or proof of consistency is claimed.

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1 The result and the comparison standard

The source repository was inspected at the immutable revision

`ae3007ca4467011e0e051c84acb4a1dd13971b0e`

Its current catalogue contains eight wave-3 packages, numbered 16–23; earlier waves survive in history and in copied references. We compare against that catalogue, rather than an older thirteen-variable starting point [1]. A smaller equation can mean fewer unknowns, lower degree, fewer operations, smaller coefficient height, or a simpler proof. These criteria disagree. Table 1 records the improvements relevant to the present construction without collapsing them into one ordering.

Construction and domain	Unknowns	Equations	Exact degree	Operations
Package 20 product, $\mathbb{Z}_{>0}$	12	1	1145	123
Package 21 product, $\mathbb{Z}_{>0}$	12	1	1353	116
Package 21 system, $\mathbb{Z}_{>0}$	12	3	1344	112
Package 22 system, $\mathbb{Z}_{>0}$	12	3	1520	112
New product, $\mathbb{Z}_{>0}$	**12**	**1**	**1097**	**116**
New system, $\mathbb{Z}_{>0}$	**12**	**3**	**1088**	**110**
New sum of squares, $\mathbb{Z}_{>0}$	12	1	2176	118
Package 20 quartic, $\mathbb{Z}$	54	1	4	–
New quartic, $\mathbb{Z}$	**51**	**1**	**4**	–
Package 20 sextic, $\mathbb{Z}$	45	1	6	–
New sextic, $\mathbb{Z}$	**42**	**1**	**6**	–
Package 20 octic, $\mathbb{Z}$	39	1	8	–
New octic, $\mathbb{Z}$	**36**	**1**	**8**	–

Table 1: Degrees of systems mean maximum residual degree. Operation counts charge every binary $+, -, \times$, including multiplication by numerals; reuse and equality comparisons are free. A dash means no optimized arithmetic operation count is asserted.

The new product lowers the best reviewed twelve-variable degree by 48 and uses seven fewer operations than that degree-minimizing predecessor. Against the old 116-operation product, it lowers degree by 256. The 110-operation system uses a shifted transport coordinate and compares the two sides of the last norm directly; these implementation choices are disclosed in Section 8. Thus the count is not obtained by silently changing the operation model.

Several existing forms remain preferable under other criteria. The explicit nine-natural-variable Jones specialization in package 18 still uses fewer unknowns than the moderate-degree construction. Its degree is $161844 \cdot 5^{33} + 484$ and its delivered circuit uses 20 531 operations. The older 67-operation product still has the smallest reported single-equation operation count. The 59-operation system in sixteen positive variables is also unaffected. No claim below beats those counts [1, 9].

For unrestricted integers, Section 9 gives a fully specified seventeen-variable equation. Its exact degree is $424296 \cdot 5^{33} + 53824$ and its circuit has 20 607 binary operations. It improves the concrete 26-integer count in the inspected direct family. It is not a global variable-count record: Sun's 2026 representation theorem gives a general ten-integer bound [11]. This article supplies no concrete ten-integer specialization and makes no claim of absolute minimality.

Meaning of the logical assertion. Fix the syntactic proof predicate used by the K215 endpoint. Writing $\mathrm{Ref}_{\mathrm{ZFC}}$ for existence of a finite ZFC derivation of falsity, the theorem asserted for each construction is

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{x} \in \mathcal{D} :\ P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}.$$

Negation gives the requested equivalence with Con(ZFC). This does not prove that either side of the negated equivalence holds. No additional claim that ZFC cannot prove inconsistency is made under mere consistency. The domain $\mathcal{D}$ is part of each theorem.

2 Fixed coefficients and inherited mathematical dependencies

The twelve-variable and low-degree constructions use the complete `wang943-compact-5pow10` tuple from package 19 [3]. Put

$$d_* = 5^{10} = 9\,765\,625, \qquad B = 2^{d_*}, \qquad \beta = B - 1, \quad K_0 = D_C + BD_R.$$

The remaining coefficients are M_C and the *shifted* source mask $M_F = M_F^{\text{native}} + B - 1$. In particular the source port M_F must not be replaced by the native mask. Every coefficient is fixed before any unknown is quantified. No coefficient depends on whether ZFC is consistent.

The archive includes the literal K215 transition table, the finite coefficient generators, the materialized coefficient payloads, and their hashes. The coefficient sizes are:

Port	Bits	Uncompressed bytes
β	9 765 625	1 220 704
K_0	13 201 251	1 650 157
M_C	9 765 625	1 220 704
M_F	9 765 626	1 220 704

The four deterministic gzip payloads total 20 260 bytes. This is a storage fact, not a small coefficient-height claim. All four reproduced raw SHA-256 digests agree with the pinned manifest.

For precision, the bounded recipe uses $\rho = 2^{625}$ and a cell length $L_c = 15625$. Its 943 ordered directed-signal Wang tiles are generated from all 430 transitions of K215. Selector 0 is the true blank-start tile; selector 1 is a globally impossible tile. Binary occupied edge labels produce forty copy positions and four anchors. If $\mathcal{E}$ is the resulting 989-position support and c_e its explicitly generated clause coefficients, the compact geometry is

$$H_c = 5497, \quad T_1 = 7631, \quad T_2 = 8733, \quad \text{anchors } 1095, 1096, 2195, 3296.$$

For $\kappa = 1024$, the clause mask is

$$\mu = \kappa - 2 + \sum_{j=1}^{44} (\kappa - 3)\kappa^j + 2^{450}(2^{78} - 1).$$

The integers are fixed by

$$\begin{aligned} D_C &= \rho^{30} + \rho^{5517} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e(\rho^{7631-e} + \rho^{8733-e}), \\ D_R &= \rho^{5497}, \\ M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \end{aligned}$$

$$M_F^{\text{native}} = \mu(\rho^{7631} + \rho^{8733}) + \sum_{j=0}^{9} (\rho^{1013+j} + \rho^{6520+j}) + \rho^{2195} + \rho^{3296} + 4.$$

The complete finite rule for $\mathcal{E}, c_e$ and the tile ordering is in `coefficients/build_wang943.py`; the compact geometric substitution is in `build_wang943_compact.py`. Thus this is a particular integer

polynomial, not a universal polynomial with an unspecified program index. The accompanying source proof establishes

$$0 < M_C, M_F^{\mathrm{native}} < B - 1, \quad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{native}} \equiv 4 \pmod 8,$$

and $\mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{native}}) = d_*$.

2.1 What is imported and what is proved here

The logical chain has three components. First, the literal blank run of K215 halts if and only if ZFC has a finite refutation. The pinned Lean declaration `KEndpoint215.blank_halts_iff_zfc_refutation` states that calibration; its source is included for identification [2]. Second, the native-window/Pell compiler translates that blank halting predicate into the arithmetic interface recalled below. Its ordinary mathematical proof, including the Wang adapter, is an inherited dependency. Third, the transformations proved in this article replace that interface by the displayed polynomials while preserving existence of solutions.

The first two components are not reproved from axioms here. Relevant source proofs are copied into `references/` and `coefficients/`. The new equivalence proof is therefore modular: it states and uses the precise existing interface, rather than treating a finite circuit test as proof of the compiler. No Lean or Rocq build of this new development was run. The assertion of provability in ZFC comes from formalizing the ordinary finite arithmetic and Pell arguments in ZFC, not from an assumed implication from Lean's metatheoretic soundness to ZFC.

3 The twelve-variable polynomial

All twelve supplied unknowns are strictly positive integers:

$$(J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r) \in \mathbb{Z}_{>0}^{12}.$$

Every other symbol in this section is a polynomial abbreviation, not an additional unknown. Define

$$\begin{array}{lllll} q = \beta J + 1, & C = q - F - \alpha, & X = wq, & Y = sq^3, & E = XY, \\ k = \eta + \zeta, & c = kY + \eta, & a = E + Y, & H = 4a + 3, & \Delta = a^2 + H, \\ \ell = EkY, & \mathcal{A} = 1 + \ell(\ell + k), & G = 1 + \Delta c^2. & & \end{array}$$

The packed index and the two outer factors are

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \tag{1}$$
$$N_k = k - (q + h)E - R, \tag{2}$$
$$N_t = (K_0 + w)C + q - F - z(q - 1). \tag{3}$$

The term $q + h$ replaces the old positive index quotient. This is an actual coordinate restriction whose completeness will be proved, not an identity on arbitrary old witnesses.

Define the strong and auxiliary expressions

$$\begin{array}{llll} I = ic^2, & D = 1 + \Delta I^2, & Q = (\Delta I)^2, & d = Q - 1, \\ v = cT, & b = c + RD, & e = Q(Dv^2 + b^2) - 1, & B_a = 2Qvb. \end{array}$$

The new, much shorter separation bound is

$$\boxed{W = I(T + I), \qquad L = 1 + HW, \qquad S = X + ac + Hr.} \tag{4}$$

Temporarily use formal square roots $u^2 = G$, $t^2 = \mathcal{A}$, and $f^2 = D$. Define

$$\mathcal{N} = \prod_{\sigma,\tau,\upsilon\in\{-1,1\}} \left[d(S - \sigma Lu - \tau HLt)^2 - H^2 e + \upsilon H^2 B_a f \right]. \tag{5}$$

The product is invariant under the three formal sign changes, so every square root cancels and $\mathcal{N}$ is a polynomial with integer coefficients in the twelve supplied variables. This statement holds even when the specialized radicands have dependent square classes or are squares. The radicals in (5) are notation for a finite polynomial identity; there are no hidden existential root variables.

The requested single equation is

$$\boxed{P_{12} = N_k N_t (1 + \mathcal{N}) - 1 = 0.} \tag{6}$$

Section 7 gives a radical-free evaluation of $\mathcal{N}$. The file `norm12/shifted_norm12_product.json` lists every one of its 116 binary operations.

Theorem 3.1. *With the fixed coefficient tuple of Section 2, ZFC proves that (6) has a positive integer solution if and only if K215 halts on its blank tape, and hence if and only if ZFC has a finite refutation. Equivalently,*

$$\mathrm{ZFC} \vdash \left(\nexists \mathbf{x} \in \mathbb{Z}_{>0}^{12} : P_{12}(\mathbf{x}) = 0 \right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The polynomial has exact degree 1097 *and a* 116*-operation circuit.*

4 Bounds established before any root is recovered

The parity argument in Section 7.1 shows that a zero of P_{12} satisfies $\mathcal{N} = 0$ and $N_k = N_t = \epsilon$ for one $\epsilon \in \{-1, 1\}$. We first derive all inequalities used by the new root bound from these outer unit equations alone.

If $C \le -1$, then

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \le 0$$

and $N_t \le -(K_0 + w) \le -2$, a contradiction. Thus $C \ge 0$; consequently $q - F = C + \alpha \ge 1$ and $1 \le F \le q - 1$. Let $G_0 = (q-1)(q-F) + \alpha = q\alpha + (q-1)C$. Then

$$q \le G_0 \le q(q-1), \qquad 0 < (M_C + qM_F)J < (1+2q)(q-1).$$

It follows directly from (1) that

$$q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2. \tag{7}$$

Also $q = \beta J + 1 \ge B > 16$, $R > 3q$, $Y \ge q^3$, and $E \ge q^4 > R + 2$. The shifted quotient $q + h$ is positive independently of the value of R.

The useful effect of the shift is the inequality $c > R(a+5)$. Indeed

$$(q+1)EY - q^4(E+Y) = qE(Y - q^3) + Y(E - q^4) \ge 0.$$

Since $R < q^4$ and $a = E + Y$, this gives $(q+1)EY > Ra$. From $N_k = \epsilon$ and $h \ge 1$ we obtain

$$c = (q+h)EY + (R+\epsilon)Y + \eta > Ra + (R-1)Y > R(a+5),$$

where the last step uses $R \ge 2$ and $Y \ge 4096$. As $a \ge 5$, we have $\Delta = a^2 + 4a + 3 < a(a+5)$. Therefore

$$\boxed{ac > R\Delta, \qquad c > a, \qquad c > R.} \tag{8}$$

These facts do not assume that u, t, f, or an auxiliary root is rational. This order is essential: using a Pell classification to justify (8) here would be circular.

5 The compact separation bound and recovery of all roots

If $\mathcal{N} = 0$, choose one vanishing factor in (5) and absorb its three signs into coherent choices of u, t, f. Set $y = (S - Lu - HLt)/H$. Then

$$\begin{aligned} u^2 &= G, & t^2 &= \mathcal{A}, & f^2 &= D, \\ dy^2 &= e - B_a f = Q(vf - b)^2 - 1, & S &= Lu + HLt + Hy. \end{aligned} \tag{9}$$

Work in a finite normal extension of $\mathbb{Q}$ containing these roots. Every conjugate y' has an associated $f' = \pm\sqrt{D}$. Hence

$$|y'| < \sqrt{Q/d}\,(v\sqrt{D} + b). \tag{10}$$

This includes complex conjugates. To see it explicitly, put $M = v\sqrt{D} + b > 1$. The real number $(vf' - b)^2$ is between 0 and M^2, so the absolute value of $Q(vf' - b)^2 - 1$ is at most $\max(1, QM^2 - 1) < QM^2$.

Lemma 5.1 (The two exact square comparisons)**.** *Under the outer unit equations and positive supplied coordinates,*

$$\sqrt{Q/d}\,c\sqrt{D} < I\sqrt{G}, \qquad \sqrt{Q/d}\,b < I^2\sqrt{G}.$$

Consequently every conjugate in (9) satisfies $|y'| < I(T + I)\sqrt{G} = W\sqrt{G}$.

Proof. For the first inequality, an exact expansion gives

$$dI^2G - Qc^2D = I^2[\Delta^2(I^2 - c^2) - G] > 0.$$

Indeed $I \geq c^2$, $c \geq 2$, and $\Delta \geq 2$, so the bracket is at least $\Delta(\Delta - 1)c^2 - 1 > 0$.

For the second inequality, (8) and integrality give $ac - R\Delta \geq 1$. Also $c + R < 2c \leq c^4 \leq I^2$. Thus

$$b = c + R + R\Delta I^2 < acI^2.$$

The remaining comparison is

$$dG - Qa^2c^2 = Q(1 + Hc^2) - G = (Q - 1) + (QH - \Delta)c^2 > 0.$$

Therefore $\sqrt{Q/d}\,ac < \sqrt{G}$, proving the second inequality. Multiplying the first by $T > 0$ and adding the second in (10) gives the stated bound. □

Because $L = 1 + HW$, the lemma implies

$$H|y'| < L\sqrt{G}. \tag{11}$$

Moreover $\sqrt{\mathcal{A}} > \ell = kEY$, while $\sqrt{G} < (a + 2)c < 2(a + 2)kY$. Since $HE > 4(a + 2)$,

$$H\sqrt{\mathcal{A}} > 2\sqrt{G}. \tag{12}$$

If t were irrational, an automorphism of the normal extension would send t to $-t$. Subtract its image of the last equality in (9). The changed HLt term has magnitude $2HL\sqrt{\mathcal{A}}$, whereas the combined magnitude of the other changed terms is less than $4L\sqrt{G}$, by (11). This contradicts (12). Thus t is rational, and therefore integral because its square is an integer.

Now an automorphism negating an irrational u would fix t. The same subtraction would give $2L\sqrt{G} < 2L\sqrt{G}$, again impossible. Thus u is an integer. Equation (9) makes y rational. Since $B_a > 0$, the equality $f = (e - dy^2)/B_a$ makes f rational and then integral. The last integrality step for y needs a special Pell lemma; rationality alone is insufficient.

Lemma 5.2 (Rational auxiliary Pell roots)**.** *If $m \geq 2$ and V are integers, $y \in \mathbb{Q}$, and $m^2V^2-(m^2-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

Proof. Write $y = A_1/B_1$ in lowest terms with $B_1 > 0$. The equation gives $B_1^2 \mid m^2 - 1$. Put $D_1 = (m^2 - 1)/B_1^2$. This is a positive nonsquare integer. The two positive Pell solutions are (m, B_1) and $(m|V|, |A_1|)$; neither ordinate is zero. If $\chi_j + \psi_j\sqrt{D_1}$ denotes the jth positive power of the fundamental positive Pell unit with integer coefficients in $\mathbb{Z}[\sqrt{D_1}]$, write these solutions as (χ_j, ψ_j) and (χ_l, ψ_l).

The elementary divisibility fact $\chi_j \mid \chi_l$ forces l to be an odd multiple of j. To verify it, the Pell addition formulas give $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$ and reflection reduces the index modulo $2j$ into $0 \leq n \leq j$. For $n < j$, strict growth gives $0 < \chi_n < \chi_j$. The only divisible case is $n = j$. Pell multiplication then gives $\psi_j \mid \psi_l$, so $B_1 \mid A_1$. Coprimality forces $B_1 = 1$. □

Apply the lemma to $m = \Delta I$ and $V = vf - b$. It proves that y is integral. Finally $L \equiv 1 \pmod{H}$ in (9) gives $u \equiv X + ac \pmod{H}$, so

$$\gamma = (u - X - ac)/H \in \mathbb{Z}.$$

The restored coordinates satisfy the first, main, strong, and auxiliary Pell equations of the inherited signed normalized interface:

$$\begin{aligned} t^2 - \ell(\ell + k) &= 1, \\ (X + ac + H\gamma)^2 - \Delta c^2 &= 1, \\ f^2 - \Delta i^2 c^4 &= 1, \\ Q((c(Tf - 1) - Rf^2)^2 - y^2) + y^2 &= 1. \end{aligned}$$

The auxiliary identity uses $f^2 = D$. The inherited signed-rank theorem resolves the common outer unit sign to $+1$ and decodes an actual blank halting computation. Signed recovered roots and γ are permitted by that interface; no positivity of those recovered coordinates is assumed [4, 5].

6 Completeness of the quotient shift

It remains to prove that imposing $h_{\text{old}} = q + h$ loses no accepting witnesses. Take a canonical positive solution of the old blank compiler. The existing first-Pell classification gives

$$k = 2\psi_{P_0}(n), \qquad P_0 = 2EY + 1, \qquad 2n = R + 1.$$

Here $\psi_P(0) = 0$, $\psi_P(1) = 1$, and $\psi_P(n + 2) = 2P\psi_P(n + 1) - \psi_P(n)$. The equality $2n = R + 1$ is a conclusion of the complete old signed-rank argument, not a premise used in Section 4 [5, Section 4]. As $R > 3q$, we have $n \geq 3$, whence

$$k \geq 2\psi_{P_0}(3) = 8P_0^2 - 2 > 32E^2Y^2.$$

The old index equation gives

$$h_{\text{old}} = (k - R - 1)/E > R > q,$$

using $R + 1 < E$ and $R < E$. Therefore $h = h_{\text{old}} - q$ is a positive integer.

For the canonical positive roots u, t, f, y and γ, define

$$r = \gamma + Wu + Lt + y > 0.$$

The identity $u = X + ac + H\gamma$ gives $S = Lu + HLt + Hy$. One factor of (5) vanishes, so $\mathcal{N} = 0$ and $P_{12} = 0$. This proves completeness. Together with the soundness argument above and the calibration of K215 it proves the semantic part of Theorem 3.1.

7 A radical-free circuit, the parity guard, and exact degree

For an explicit finite formula with no displayed radicals, use pairs (x_0, x_1) representing $x_0 + x_1 u$ in $\mathbb{Z}[\mathbf{x}][u]/(u^2 - G)$. Pair multiplication is

$$(x_0, x_1)(y_0, y_1) = (x_0 y_0 + G x_1 y_1,\ x_0 y_1 + x_1 y_0).$$

All pair operations below are this fixed finite arithmetic rule. Put

$$\begin{aligned} A_0 &= H^2 L^2 \mathcal{A}, & V &= (S^2 + L^2 G,\ 2SL), \\ \mathcal{C} = C_0 + C_1 u &= d(V + A_0) - H^2 e, & K &= 4d^2 A_0 V, \\ Z_0 &= H^4 B_a^2 D, & N_0 + N_1 u &= (\mathcal{C}^2 - K - Z_0)^2 - 4KZ_0. \end{aligned}$$

Scalars such as $A_0, H^2 e, Z_0$ are inserted as pairs with second coordinate zero. Then

$$\mathcal{N} = N_0^2 - G N_1^2. \tag{13}$$

Expanding the two conjugates in t and the two in f verifies this identity with (5). The identity

$$(\mathcal{C}^2 + K - Z_0)^2 - 4\mathcal{C}^2 K = (\mathcal{C}^2 - K - Z_0)^2 - 4KZ_0$$

avoids a full pair multiplication because Z_0 is scalar. The circuit also uses $Z_0 = 4H^4(QDv^2)(Qb^2)$, reusing products already needed for e. These optimizations are inherited from package 21; the new changes are the quotient shift and smaller bound.

7.1 Why the product really enforces all equations

The coefficient of u in V is even; hence C_1 is even, the odd coefficient of $\mathcal{C}^2$ is divisible by 4, and that of K by 8. The odd coefficient of $(\mathcal{C}^2 - K - Z_0)^2$ is therefore divisible by 8, as is that of $4KZ_0$. Thus $8 \mid N_1$ as an integer polynomial and

$$\mathcal{N} \equiv N_0^2 \pmod{64}.$$

In particular $\mathcal{N} \neq -2$. If $N_k N_t (1 + \mathcal{N}) = 1$, all three factors are integer units. The last cannot be -1, so it is 1, $\mathcal{N} = 0$, and $N_k = N_t$ are common units. This proves the assertion used at the start of Section 4, without any sign or root assumption.

7.2 Uniform exact degree

Treat the four fixed ports as degree zero and every supplied unknown as degree one. The important degrees are

Expression	Degree	Expression	Degree	Expression	Degree
c	5	a, H, E	6	Δ	12
$\mathcal{A}, G$	22	I	11	D	34
Q, d	46	b	38	e	122
W	22	L	28	S	11
A_0	90	C_0	136	C_1	85
N_0	544	N_1	≤ 493	$\mathcal{N}$	1088

The highest term of C_0 comes uniquely from dA_0: the competing $H^2 e$ has degree 134 and dV_0 degree 124. Therefore the leading term of $\mathcal{N}$ comes from C_0^8 and cannot cancel against GN_1^2, whose degree is at most 1008.

For a uniform certificate put $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The leading homogeneous form is

$$\mathcal{N}_{\mathrm{top}} = 2^{64} i^{48} k_0^{112} w^{80} s^{192} q_0^{656}.$$

The leading factors of N_k and N_t are respectively

$$-(q_0 + h) w s q_0^4, \qquad w(q_0 - F - \alpha) - z q_0.$$

Thus $\deg P_{12} = 1088 + 7 + 2 = 1097$. For example the coefficient of

$$J^{661} h z\, i^{48} \eta^{112} w^{81} s^{193}$$

is $2^{64}\beta^{661} > 0$. This proves exact degree for the authentic coefficient tuple, not just for a small numerical specialization.

8 The 110-operation system and domain conventions

The unshifted three-equation presentation is $N_k = 1$, $N_t = 1$, and $\mathcal{N} = 0$. Its most economical delivered comparison form uses 111 operations, since (13) can be checked as $N_0^2 = G N_1^2$ without calculating their difference.

For one further saving substitute $z_{\mathrm{old}} = z + 1$ in the transport equation. The result is

$$\boxed{N_k = 1, \qquad (K_0 + w)C = F + z(q-1), \qquad N_0^2 = G N_1^2.} \tag{14}$$

The norm is independent of z. The exact identity underlying the shift is

$$(K_0 + w)C + q - F - (z+1)(q-1) - 1 = (K_0 + w)C - F - z(q-1).$$

Every positive shifted solution gives a positive unshifted solution. Conversely a canonical old witness has $C > 0$, $X = wq = 2^R$, and $R > 3q$, so $w > q > F$. The old transport equation then forces $(z_{\mathrm{old}} - 1)(q-1) = (K_0 + w)C - F > 0$; hence $z_{\mathrm{old}} \geq 2$. The shift is complete.

The exact operation ledgers are

Form	Multiplications	Additions/subtractions	Total
Product P_{12}	73	43	116
System (14)	71	39	110
Sum of the three squared residuals	74	44	118

All gates and all supplied ports are live. The sum of squares has degree 2176, since $\mathcal{N}^2$ is its unique highest-degree summand. The product uses the old transport coordinate; the system and its sum of squares use the shifted one. They have equivalent solvability predicates, not identical named-coordinate zero sets.

For a formulation over $\mathbb{N} = \{0, 1, 2, \ldots\}$, replace each of the twelve positive inputs by one plus a new natural input. This preserves variable count and degree and adds at most twelve input additions to the literal circuit. Reading the displayed positive-domain polynomial directly over all integers is not justified. No 26-integer extension of the new compact bound is asserted.

9 A concrete equation in seventeen unrestricted integers

This section gives a fixed polynomial P_{17} in seventeen unrestricted integer unknowns whose absence of zeros is equivalent, provably in ZFC, to Con(ZFC). It uses the complete original T4 coefficient tuple of package 18, `18-jones9`, at revision `ae3007ca4467011e0e051c84acb4a1dd13971b0e` [9]. All source coefficients and abbreviations below are local to this section. In particular, this construction has its own coefficient binding, distinct from the 943-tile binding used in the preceding sections.

The result improves the delivered concrete 26-integer count in the inspected direct compiler family, at the cost of an enormous degree. It establishes neither global minimality nor a universal-variable record: Sun's ten-integer representation theorem already gives a smaller abstract general bound [11]. The contribution here is the complete fixed specialization, its explicit circuit, and the additional proofs needed to use signed coordinates. The source Jones ratio theorem, quartic/compiler theorem, and K215 calibration are inherited ordinary mathematical theorems. The new arguments below are proved in full; no new proof-assistant certificate is claimed.

9.1 The fixed coefficients and exact source interface

Start with eight base coordinates

$$(\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0) \in \mathbb{Z}^8$$

and abbreviate

$$g = g_0 + 1, \quad h = h_0 + 1, \quad s = s_0 + 1, \quad w = w_0 + 1, \quad i = i_0 + 1, \quad j = j_0 + 1.$$

The old ninth coordinate of package 18 and its finalizer are replaced below. The full source expressions through the ratio conditions are reproduced here so that the signed-domain argument has an explicit interface.

Finite coefficient recipe. Put $\nu = 32$ and $L = 5^{33}$. Shift the 32 positive coordinates of the pinned quartic by 1, and include its unused dummy input as coordinate zero. The dummy is fixed to 1 and is not an unknown of P_{17}. Let p_α be the resulting coefficient indexed by $\alpha \in \mathbb{N}^{33}$, with $|\alpha| = \sum_{r=0}^{32} \alpha_r \le 4$, and put

$$c_\alpha = (4 - |\alpha|)! \left(\prod_{r=0}^{32} \alpha_r! \right) p_\alpha.$$

The four compiler ports are bound to the complete original T4 tuple in `signed17/upstream/COEFFICIENT_BINDINGS.json`. The coefficient records and the positive-coefficient bounds ρ_j for the shifted source residuals are specified by `signed17/upstream/nine/generated/constants.json` and `signed17/upstream/nine/generated/index_bound_s.json`. Define the following fixed integers and fixed polynomial:

$$s_\star = 2 + 24 \sum_j \rho_j^2, \qquad z_\star = 2^{s_\star},$$

$$e_\star(T) = \sum_{\substack{\alpha \in \mathbb{N}^{33} \\ |\alpha| \le 4}} (z_\star + c_\alpha) T^{L - \sum_{r=0}^{32} \alpha_r 5^r}, \qquad y_\star = e_\star(2z_\star).$$

The sum includes the unlisted zero coefficients. All exponents are fixed nonnegative integers. The source bound proves

$$z_\star > |c_\alpha| + 1 \quad \text{for every } \alpha, \qquad y_\star \ge 2z_\star.$$

Thus the coefficient recipe is fixed independently of the quantified unknowns and of the consistency of ZFC.

Source polynomial abbreviations. Define

$$b = y_\star + 1 + \varepsilon, \qquad \mathcal{B} = b^5, \qquad q = \mathcal{B}^L,$$
$$\ell = \sum_{r=1}^{\nu} \mathcal{B}^{5^r}, \qquad \lambda = \sum_{r=0}^{4L-1} \mathcal{B}^r, \qquad e = e_\star(\mathcal{B}),$$
$$\theta = \mathcal{B} - 2z_\star, \qquad N = q^{16}, \qquad G = 32b\mathcal{B}^{5^\nu}.$$

To display the packed source expression without a long line, write

$$\begin{aligned} R_1 &= g + eq^3 + \ell q^5 \\ &\quad + \left[2(e - z_\star\lambda)(1 + \mathcal{B} + g)^4 + \lambda\mathcal{B}(1 + q^4)\right]q^7, \\ R_2 &= q^3 - b\ell + \ell + \theta\lambda q^3 + (\mathcal{B} - 2)q^8, \\ R &= R_1(N^2 - N) + R_2(N^2 - 1). \end{aligned}$$

The remaining abbreviations are

$$U = N^2 w, \qquad Y = N^2 s, \qquad M = RY,$$
$$P = 2M^2 U, \qquad K = R + 1 + h(P - 1), \qquad A = M(U + 1),$$
$$B_1 = 2R + 1, \qquad W_1 = bw, \qquad C = B_1 + W_1 + \phi,$$
$$J = 4A - 5, \qquad Q = 3W_1 C - 2(W_1^2 - 1),$$

and

$$D = (A^2 - 1)C^2 + 1, \qquad E = iJDC^2,$$
$$F = (A^2 - 1)E^2 + 1, \qquad G_1 = A + F(CD + 1 - A),$$
$$H = B_1 + jC, \qquad I = (G_1^2 - 1)H^2 + 1,$$
$$a_0 = (P^2 - 1)K^2 + 1, \qquad a_1 = DFI,$$
$$D_0 = JF, \qquad T_0 = QF + J(H - C).$$

Finally, set

$$m = K^2 G - 4(C - KY)^2 G - 8gK^2.$$

Let $\mathrm{Sq}(\mathbb{Z}) = \{r^2 : r \in \mathbb{Z}\}$. The imported Jones ratio equivalence states that there exist eight *nonnegative* base coordinates satisfying

$$a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}), \qquad D_0 \mid T_0, \qquad m > 0 \tag{15}$$

if and only if the pinned 32-variable quartic has a positive solution. The imported compiler and K215 calibration identify that assertion with existence of a finite ZFC refutation. These are the precise inherited theorem boundaries used below. The signed-coordinate argument depends on the displayed source structure, not merely on existence of an unspecified nine-variable representation.

9.2 Two nonnegativity guards are redundant

Only six base coordinates need explicit nonnegativity guards:

$$\mathbf{x} = (x_1, \ldots, x_6) = (\varepsilon, g_0, s_0, w_0, \phi, j_0).$$

Lemma 9.1 (The source denominator never vanishes)**.** *For every integer assignment to the eight base coordinates, $D_0 \neq 0$.*

Proof. The integer $J = 4A - 5$ cannot vanish. If $F = 0$, then $(A^2 - 1)E^2 = -1$, which forces $A = 0$ and $E = \pm 1$. But $A = 0$ gives $J = -5$, so $E = iJDC^2$ is divisible by 5, a contradiction. □

Lemma 9.2 (A signed coordinate)**.** *If $i \neq 0$, replacing i by $|i|$ leaves a_0, a_1, D_0, T_0, m unchanged.*

Proof. The coordinate i enters through E, and these five expressions use E only through E^2 in F. The replacement therefore preserves all five values. The new base coordinate $i_0 = |i| - 1$ is nonnegative. □

Lemma 9.3 (The guard on h_0 follows from the source conditions)**.** *Suppose that the six coordinates* $\mathbf{x}$ *are nonnegative, $a_0 \in \mathrm{Sq}(\mathbb{Z})$, and $m > 0$. Then $h \geq 1$, so $h_0 \geq 0$.*

Proof. The coefficient recipe gives $b \geq 2$, $\mathcal{B} = b^5 \geq 32$, $e \geq 0$, and $\mathcal{B} > 2z_\star$. Put $t = 5^\nu$. In particular, $G > 0$ and $g \geq 1$. The positive margin first implies

$$K \neq 0, \qquad g < G/8 = 4b\mathcal{B}^t, \qquad 4(C - KY)^2 < K^2. \tag{16}$$

Indeed, $K = 0$ would make $m \leq 0$, while $m \leq (G - 8g)K^2$ gives the bound on g. Deleting the strictly positive term $8gK^2$ from the margin inequality and dividing by G gives the last inequality.

Since $b \leq \mathcal{B}$ and $\mathcal{B} > 5$, we have

$$\begin{aligned} 1 + \mathcal{B} + g &< 1 + \mathcal{B} + 4\mathcal{B}^{t+1} \leq 5\mathcal{B}^{t+1} \\ &< \mathcal{B}^{t+2} \leq \mathcal{B}^{5t} = q. \end{aligned} \tag{17}$$

The distinct exponents in ℓ lie between 0 and t, so

$$\ell \leq \sum_{r=0}^{t} \mathcal{B}^r < \mathcal{B}^{t+1}, \qquad b\ell < \mathcal{B}^{t+2} \leq q < q^3.$$

Writing $V_0 = 1 + \mathcal{B} + g$, the possibly negative part of R_1 is dominated by its positive companion:

$$\lambda\mathcal{B}q^4 - 2z_\star\lambda V_0^4 > 0.$$

This follows from (17) and $\mathcal{B} > 2z_\star$. After separating $2(e - z_\star\lambda)V_0^4$ into its two terms, all remaining terms of R_1 are nonnegative, and $g > 0$. Also $R_2 > 0$, since $q^3 > b\ell$, $\theta > 0$, and $\mathcal{B} > 2$. Consequently

$$R \geq N^2 - N \geq 2.$$

This proves positivity of R from the margin-derived bound, without assuming any sign for h.

Now $Y, U \geq 1$, $C = 2R + 1 + bw + \phi > 0$, and $P = 2R^2Y^2U \geq 2R^2 > R + 2$. If $K < 0$, then $|C - KY| = C + |K|Y > |K|$, contrary to (16). The case $K = 0$ has already been excluded, so $K > 0$.

Suppose $h \leq 0$. Then $1 \leq K \leq R + 1 < P$. If $K = 1$, the identity $K = R + 1 + h(P - 1)$ gives $P - 1 \mid R$, contrary to $0 < R < P - 1$. If $2 \leq K < P$, then

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2.$$

The difference on the left is $K(2P - K) > 0$, and the difference on the right is $K^2 - 1 > 0$. Thus a_0 lies strictly between consecutive squares, a contradiction. Hence $h \geq 1$. □

It follows that the existential predicate (15) is unchanged if h_0 and i_0 range over all integers, provided $i = i_0 + 1$ is nonzero and the six coordinates $\mathbf{x}$ remain nonnegative.

9.3 Seven inequalities from one square condition

Introduce seven integer unknowns $z_1, \dots, z_7$. To avoid confusing the following factors with the source quantities B_1 and C, use the symbols η_j and Γ_j. Define

$$\begin{aligned} &\Gamma_0 = 1, \\ &d_j = 3x_j + 2 \qquad\qquad (1 \le j \le 6), \\ &d_7 = 3m - 1, \\ &\eta_j = 1 + 4d_j\Gamma_{j-1}^2 z_j^2, \quad \Gamma_j = \Gamma_{j-1}\eta_j \qquad (1 \le j \le 7). \end{aligned} \tag{18}$$

Lemma 9.4 (A coprime Pell stack)**.** *Subject to $z_1 \cdots z_7 \neq 0$, the condition $\Gamma_7 \in \mathrm{Sq}(\mathbb{Z})$ implies $x_1, \dots, x_6 \ge 0$ and $m > 0$. Conversely, if these seven inequalities hold, there are nonzero $z_1, \dots, z_7 \in \mathbb{Z}$ for which Γ_7 is a square.*

Proof. Every η_j is 1 modulo 4, hence nonzero, and $\gcd(\eta_j, \Gamma_{j-1}) = 1$. Thus the seven factors η_j are pairwise coprime. If their product is a square, unique factorization implies that each $|\eta_j|$ is a square. The negative of an integer square is 0 or 3 modulo 4, so $\eta_j \equiv 1 \pmod 4$ forces each η_j to be a positive square.

Both Γ_{j-1} and z_j are nonzero. If $d_j < 0$, then $\eta_j \le 1 - 4 = -3$, a contradiction. Hence $d_j \ge 0$. Since $d_j \equiv 2 \pmod 3$, equality is impossible. Therefore $d_j > 0$, which is exactly $x_j \ge 0$ for $1 \le j \le 6$ and $m \ge 1$ for $j = 7$.

Conversely, every d_j is then positive and congruent to 2 modulo 3, so it is not a square. The Pell equation $u_j^2 - d_j v_j^2 = 1$ has a nontrivial positive solution. It has one whose v_j is divisible by any prescribed positive integer M_0: powers of the associated unit matrix

$$\begin{pmatrix} u_j & d_j v_j \\ v_j & u_j \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$$

eventually become the identity modulo M_0, because $\mathrm{SL}_2(\mathbb{Z}/M_0\mathbb{Z})$ is finite. A positive power remains nontrivial and has positive lower-left entry. Set $M_0 = 2|\Gamma_{j-1}|$ and choose such a solution. Then

$$z_j = \frac{v_j}{2\Gamma_{j-1}} \in \mathbb{Z} \setminus \{0\}, \qquad \eta_j = u_j^2.$$

Induction gives seven positive-square factors and a square product. □

The nonzero conditions on the seven witnesses will be imposed together by one divisibility relation. In particular, no zero witness is being excluded without an algebraic constraint.

9.4 Divisibility and nonvanishing through a degree-five polynomial

Let

$$\Pi(t) = (t^3 - t - 1)(t^2 + 23).$$

Lemma 9.5 (Intersectivity)**.** *The polynomial Π has no rational root, but for every nonzero integer n there is an integer t such that $n \mid \Pi(t)$.*

Proof. The monic cubic $f(t) = t^3 - t - 1$ has no rational root: the only candidates are 1 and -1, and neither is a root. The quadratic has no real root.

It remains to construct a root modulo every prime power. The cubic has discriminant -23. For an odd prime $p \neq 23$, any root of the cubic modulo p is simple and therefore lifts to all powers of p. If it has no root modulo p, it is irreducible of degree three. Its roots lie in $\mathbb{F}_{p^3}$, and Frobenius permutes them by a three-cycle. The product of their pairwise differences is fixed by this even permutation, so lies in

$\mathbb{F}_p$; its square is -23. It is nonzero. Thus the quadratic t^2+23 has a simple root modulo p, which also lifts to every power of p. For $p=23$, the cubic has the simple root 3, because $f(3)=23$ and $f'(3)=26$.

For $p=2$, use $-23\equiv 1 \pmod 8$. If an odd t satisfies $t^2\equiv -23 \pmod{2^e}$ with $e\geq 3$, then one of t and $t+2^{e-1}$ satisfies the same congruence modulo 2^{e+1}: their squares differ by $2^e t$ modulo 2^{e+1}. The initial cases modulo 2, 4, and 8 are immediate. This supplies roots modulo every power of 2. The Chinese remainder theorem now gives a root modulo $|n|$. □

For integer arguments d,t,v, define

$$\begin{aligned}\mathcal{R}(d,t,v) = \big((t+dv)^3 - d^2(t+dv) - d^3\big) \\ \cdot \big((t+dv)^2 + 23d^2\big).\end{aligned} \tag{19}$$

This is homogeneous of degree five in d and t, with v treated as a parameter.

Lemma 9.6 (Combining divisibility with nonvanishing)**.** *For integers d,t,z with $d\neq 0$,*

$$(d \mid t \ \wedge\ z\neq 0) \quad\Longleftrightarrow\quad \exists v\in\mathbb{Z}:\ z d^5 \mid \mathcal{R}(d,t,v). \tag{20}$$

Moreover, $\mathcal{R}(d,t,v)\neq 0$ for every integer v.

Proof. The identity $\mathcal{R}(d,t,v) = d^5\Pi(t/d+v)$ and Lemma 9.5 prove nonvanishing. If $t=dq_0$ and $z\neq 0$, choose v such that $z\mid \Pi(q_0+v)$, again using Lemma 9.5. Then $zd^5\mid \mathcal{R}(d,t,v)$.

Conversely, if $z=0$, the asserted divisibility would mean $\mathcal{R}(d,t,v)=0$, which is impossible. If $d\nmid t$, choose a prime p with $\alpha = v_p(d) > \beta = v_p(t)$; here $t\neq 0$. For $r=t+dv$ we have $v_p(r)=\beta$. In $r^3-d^2r-d^3$, the first term has the unique least valuation 3β. In r^2+23d^2, the first term has the unique least valuation 2β. Hence

$$v_p(\mathcal{R}(d,t,v)) = 5\beta < 5\alpha = v_p(d^5),$$

contrary to the asserted divisibility. □

The general intersectivity mechanism is classical. The argument above supplies the degree-five instance needed here without making a novelty claim for Π itself. Sun's recent construction uses a related degree-six divisibility/nonvanishing device [11, Section 5].

9.5 The actual polynomial and its square finalizer

The quantified coordinates are exactly

$$\boldsymbol{\xi} = (\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, z_1, \dots, z_7, v, u) \in \mathbb{Z}^{17}.$$

There are eight base coordinates, seven Pell witnesses, one residue coordinate v, and one final norm coordinate u. Using (18), define

$$\begin{aligned} C_* &= \Gamma_7, & w_1 &= 1+C_*^2,\\ w_2 &= w_1(P^2+K^2+2), & Z &= (i_0+1)z_1\cdots z_7,\\ S &= ZD_0^5, & T &= \mathcal{R}(D_0,T_0,v),\\ X &= Su-T. \end{aligned}$$

Then put

$$\begin{aligned} A_0 &= S^2C_*, & A_1 &= S^2w_1^2a_0, & A_2 &= S^2w_2^2a_1,\\ V &= X^2+A_2-A_0-A_1, & W &= V^2+4X^2A_2-4A_0A_1. \end{aligned}$$

The required fixed polynomial is

$$\boxed{P_{17} = W^2 - 16V^2X^2A_2.} \tag{21}$$

Every intermediate symbol is a polynomial abbreviation, so none adds a quantified coordinate. The circuit in `signed17/generated/signed17.json` expands these abbreviations into a finite division-free directed acyclic graph.

Lemma 9.7 (The three-root finalizer)**.** *With the other sixteen coordinates fixed, the assertion*

$$\exists u \in \mathbb{Z} : \; P_{17}(\boldsymbol{\xi}) = 0$$

is equivalent to

$$S \neq 0, \qquad S \mid T, \qquad C_*, a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}). \tag{22}$$

In particular, these conditions are necessary for every zero; conversely, given these conditions, an integer u can be chosen.

Proof. Lemmas 9.1 and 9.6 give $T \neq 0$. If $S = 0$, then $A_0 = A_1 = A_2 = 0$, $X = -T$, and $P_{17} = T^8 \neq 0$. Thus a zero must have $S \neq 0$.

Choose complex roots $r_0^2 = C_*$, $r_1^2 = a_0$, and $r_2^2 = a_1$. Direct multiplication gives the polynomial identity

$$P_{17} = \prod_{\epsilon \in \{-1,1\}^3} [X - S(\epsilon_0 r_0 + \epsilon_1 w_1 r_1 + \epsilon_2 w_2 r_2)]. \tag{23}$$

One can verify it first by pairing the signs of r_0, r_1, obtaining $(Y^2 - A_0 - A_1)^2 - 4A_0A_1$, and then substituting $Y = X \pm Sw_2r_2$ and multiplying the two results. Therefore a zero makes one of the signed sums in (23) equal to the rational number $u - T/S$.

For every integer C_*, $\sqrt{|C_*|} \leq C_*^2$, including $C_* = 0$. For all real P, K,

$$\sqrt{|(P^2-1)K^2 + 1|} \leq P^2 + K^2 + 1. \tag{24}$$

Indeed, $|(P^2-1)K^2 + 1| \leq (P^2+1)K^2 + 1$, and subtraction from $(P^2 + K^2 + 1)^2$ leaves $P^4 + K^4 + P^2K^2 + 2P^2 + K^2 \geq 0$. Consequently

$$w_1 > \sqrt{|C_*|}, \qquad w_2 > \sqrt{|C_*|} + w_1\sqrt{|a_0|}. \tag{25}$$

Consider two sign patterns that differ on at least one nonzero root, and take the largest index at which they differ on a nonzero root. The absolute value of that root is at least 1, since its nonzero square is an integer. By (25), its weighted contribution to the difference strictly exceeds the sum of the absolute values of all earlier contributions. The two sums are therefore distinct. This triangle-inequality argument applies equally to negative radicands, whose chosen roots are complex.

The field generated by the three roots is a finite Galois extension of $\mathbb{Q}$. Every automorphism changes root signs. A rational encoded sum is fixed by every automorphism, so the separation just proved forces each root in that sum to be fixed individually. Every root is therefore rational. A rational number whose square is an integer is an integer; hence C_*, a_0, a_1 are nonnegative integer squares. Zero radicands cause no difficulty, because their roots are already zero and their signs can be ignored. The encoded sum is now an integer, so T/S is an integer as well. This proves necessity of (22).

Conversely, choose integer roots of the three squares and set

$$u = T/S + r_0 + w_1r_1 + w_2r_2 \in \mathbb{Z}.$$

The corresponding factor in (23) vanishes. □

9.6 Equivalence to the fixed consistency statement

Theorem 9.8 (The seventeen-integer equation)**.** *With the coefficients fixed by the recipe above,*

$$\mathrm{ZFC} \vdash \left[\left(\nexists \boldsymbol{\xi} \in \mathbb{Z}^{17} : P_{17}(\boldsymbol{\xi}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC})\right].$$

Proof. A zero gives (22). Since $D_0 \neq 0$, the condition $S \neq 0$ gives $i \neq 0$ and $z_j \neq 0$ for every j. Lemma 9.6 gives $D_0 \mid T_0$. Lemma 9.4 gives the six nonnegative base coordinates and $m > 0$. Lemma 9.3 then gives $h_0 \geq 0$. Finally, Lemma 9.2 permits replacement of i by $|i|$, yielding $i_0 \geq 0$ without changing the ratio conditions. Thus the original eight-natural-variable predicate (15) has a solution. Its imported equivalence supplies a finite ZFC refutation.

Conversely, a finite ZFC refutation supplies a source solution of (15), for which $i > 0$. Lemma 9.4 supplies nonzero $z_1, \ldots, z_7$ and a square C_*. Lemma 9.6 supplies v with $S \mid T$; also $S \neq 0$. Lemma 9.7 supplies u, giving an integer zero of P_{17}.

All new steps are elementary number theory and finite Galois theory formalizable in ZFC. Negating the proved equivalence between existence of a zero and existence of a refutation gives the assertion. □

This theorem proves the required equivalence. It does not prove either consistency or inconsistency of ZFC.

9.7 Exact complexity and reproducible evidence

The delivered circuit has exactly 20 607 binary operations: 672 additions, 26 subtractions, and 19 909 multiplications. Each multiplication by a numeral and each squaring is charged. As in package 18, construction of the fixed coefficients is outside the arithmetic evaluation count. The 348 named fixed leaves are all bound by the included coefficient recipe.

The exact total degree is

$$\deg P_{17} = 424296\,5^{33} + 53824 = 49\,394\,555\,389\,881\,134\,033\,203\,178\,824. \tag{26}$$

Thus this variable-count improvement has an enormous degree; the moderate-degree 26-integer forms retain a substantial advantage on degree and coefficient complexity.

For a degree audit, recall $L = 5^{33}$. The source degrees are

$$\deg m = 2221L + 11, \qquad \deg a_0 = 3800L + 14,$$
$$\deg a_1 = 24640L + 68, \qquad \deg D_0 = 6875L + 20.$$

The first six guard products have degree recurrence $c_j = 3c_{j-1} + 3$, with $c_0 = 0$, so $\deg \Gamma_6 = 1092$. The remaining degrees are

Expression	Exact degree
C_*	$2221L + 3289$
w_1	$4442L + 6578$
w_2	$6342L + 6586$
S	$34375L + 108$
T	$34375L + 105$
X	$34375L + 109$
A_2	$106074L + 13456$

The degree of A_2 strictly exceeds the degrees of X^2, A_0, A_1. Consequently the leading homogeneous parts of V, W, and P_{17} are respectively those of A_2, A_2^2, and A_2^4. The source and stack leading coefficients are

nonzero, giving (26). The generator also propagates structural degree bounds and a diagonal leading coefficient modulo 1 000 003. The output coefficient is 40 718, is nonzero, and is certified independent of every fixed coefficient port.

The deterministic finite diagnostics include 24 modular comparisons using the actual original T4 coefficients, 150 independent evaluations of the eight-factor norm identity, roots of Π modulo every modulus from 1 through 1200, 10 098 consecutive-square-gap checks, and the signed-i symmetry. These checks supplement the universal proofs above. Run

```
python3 -B signed17/build_signed17.py
```

from the archive root. The standard-library generator writes `signed17/generated/signed17.json` and `signed17/generated/signed17_checks.json`. The latter is a diagnostic receipt, not a proof-assistant certificate.

9.8 Provenance and remaining opportunities

The unchanged source expressions and coefficient modules come from package 18 at revision `ae3007ca4467011e0e051c84acb4a1dd13971b0e`. That package records the earlier source acquisition revision `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`. All nineteen imported files are authenticated byte for byte against their pinned Git blob identifiers, with SHA-256 hashes recorded in `signed17/SOURCE_PINS.json`. The literal quartic and ordered tile list also match the hashes in the upstream coefficient binding. The generator checks every source pin before rebuilding the polynomial. The archive preserves the upstream license and attribution files.

Several questions remain open for further simplification. Recovering the j_0 or ϕ nonnegativity condition from the other source conditions would remove another guard witness. A complete fixed instantiation of the published ten-integer construction would improve the variable count further. Optimizing the digit-polynomial circuit could reduce its roughly 20 500 inherited operations. Finally, a rebinding to the smaller 943-tile tuple would require proof of the complete quartic/compiler interface for that tuple; individual coefficient ports cannot be spliced between the two bindings. The present section makes no exhaustive novelty or absolute-minimality claim for its general devices.

10 Three fewer integer variables in the low-degree forms

Package 20 supplies exact polynomial lifts over unrestricted integers with 54, 45, and 39 variables at degrees 4, 6, and 8, respectively. Package 23 proves an independent odd-scale theorem for the signed arithmetic interface. Their composition removes three positivity parameters in each lift [6, 8].

In the source lifts, the retained scale s is forced positive by a separate residual

$$s - 1 - s_1^2 - s_2^2 - s_3(s_3 + 1).$$

Delete this residual and the three otherwise unused parameters. Replace s everywhere else by $2v_s + 1$, with $v_s \in \mathbb{Z}$. All remaining residuals are unchanged except for this affine substitution. The resulting systems have the following exact sizes:

Form	Integer variables	Residuals	Maximum residual degree	SOS degree
Q_{51}	51	30	2	4
Q_{42}	42	21	3	6
Q_{36}	36	15	4	8

The full expanded residual lists are the three JSON files under `low_degree/generated/`. Defining Q_n as the sum of their squares completely specifies an ordinary integer polynomial.

10.1 Why oddness is sufficient

The imported odd-scale theorem guards only $J, F, \alpha, z, \eta, \zeta > 0$, uses an odd strong multiplier and a square power scale, and allows the remaining coordinates to be signed. It proves that an odd nonzero s must in fact be positive at an arithmetic solution. Its negative-s branch recovers the main Pell index and obtains one of two incompatible congruences, using explicit bounds for the negative-parameter Pell sequence. This is a semantic theorem; simply deleting a positivity guard would not prove it.

For completeness the compact 943-tile tuple retains independent sufficiently large five-power tableau widths and heights. The canonical dyadic exponent $t = d_*N$ is odd, $\mathrm{pc}(R) = 3t+2$, and the half-binomial valuation is $v_2(Y) = \mathrm{pc}(R) - 2 = 3t$. Since $Y = sq^3$ and $q = 2^t$, the canonical scale s is odd. Thus every accepting tableau has the required new coordinate $v_s = (s-1)/2$. The exact population and padding hypotheses follow from the whole-tuple Wang adapter, not from splicing individual coefficients.

10.2 The exposed quotients in the quartic

The quartic lift retains quotient variables o, j in place of the single eliminated quotient T. Its equations give $V = of - c = jc - R$, hence

$$V \equiv -c \pmod{|f|}, \qquad V \equiv -R \pmod{|c|}.$$

These are precisely the congruences used by the negative-scale proof. Consequently that proof excludes $s < 0$ before any division to recover T. Once $s > 0$, the ordinary strong-rank lemma gives $f^2 \equiv 1 \pmod{c}$. Since $of + R = (j+1)c$, the identity

$$o + Rf = f(of + R) - o(f^2 - 1)$$

proves $c \mid o + Rf$. Thus $T = (o + Rf)/c$ is an integer and $c(Tf - 1) - Rf^2 = of - c = V$. The full signed interface is recovered with no unchecked divisibility assumption. The sextic and octic eliminations are already exact integer identities and require no extra quotient argument.

The full detailed composition proof is `low_degree/ODD_SCALE_LIFTS_PROOF.md`. Over $\mathbb{Z}$ a sum of squares vanishes exactly when every residual does, proving

$$\mathrm{ZFC} \vdash (\nexists \mathbf{x} \in \mathbb{Z}^n : Q_n(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}) \qquad (n = 51, 42, 36).$$

10.3 Exactness of the degrees

An affine substitution cannot increase degree. To show that degree does not fall, the exact symbolic expansion gives

$$[\tau^4]Q_{51} = 1, \qquad [J^2 L_{\mathrm{reg}}^2 v_s^2]Q_{42} = 4\beta^2, \qquad [J^6 v_s^2]Q_{36} = 4\beta^6.$$

Here L_{reg} denotes the retained register called `L` in the sextic JSON, not the norm weight from Section 3. Since $\beta > 0$, the degrees are exactly $4, 6, 8$. The sparse symbolic sum-of-squares term counts are $342, 467, 845$, respectively; these are not claims about optimally minimized circuits or numerical coefficient height.

11 Reproducibility, proof status, and integration

The archive is a self-contained research contribution with the article, complete source, emitted circuits, literal coefficient data, provenance, and verification receipts. Run

```
python3 reproduce.py
```

from its top directory. The default run uses only Python's standard library; LaTeX is needed only to rebuild this PDF. Optional full coefficient materialization is controlled by the driver. All inputs required by the default checks are included, addressing the pinned catalogue's warning that earlier verification drivers could fail after copied files were omitted during repository placement.

The performed checks separate three types of evidence. Exact symbolic and integer checks verify the new polynomial definitions: independent conjugate products, the transport shift, parity, liveness, and operation counts. Uniform leading-term proofs establish exact degree; modular leading-form fixtures are additional diagnostics. Byte-level checks verify the literal coefficient payloads against their immutable manifest. None of these tests is substituted for the semantic Pell/compiler proofs.

In particular no run purports to find an accepting K215 computation or to decide the existence of a solution. Exhaustive bounded searches could not prove the required absence of solutions. The new proofs are ordinary mathematical arguments, with the precise formalization boundary stated in Section 2.

The intended repository destination is a new report directory in the `con-zfc-diophantine-forms` collection. No source repository branch, commit, or pull request was changed. The delivery keeps the inherited sources separate from the new circuits and records one authoritative coefficient binding for each family.

12 Further research questions

1. **Can the index shift be optimized together with the norm?** The useful invariant is $ac > R\Delta$, not the particular formula $q + h$. An alternative witness parameterization that guarantees this inequality with fewer arithmetic gates might lower the operation count as well as degree. It must retain a proved completeness map.

2. **Can one remove another supplied root coordinate?** The present result keeps twelve positive inputs. Eliminating an additional coordinate requires a new integrality or congruence argument; a formal resultant alone can introduce spurious rational roots. Any proposed eleven-variable form should state the root-recovery order explicitly.

3. **Can the 67-operation polynomial be improved?** The new norm does not approach that operation count. Joint optimization of the six-factor normalized interface, including quotient substitutions and modular unit guards, remains a distinct problem.

4. **Can more positivity guards be inferred?** The odd-scale theorem removes three parameters from each low-degree lift. Identifying another coordinate whose sign is forced by the existing equations would directly improve the quartic, sextic, and octic counts.

5. **Can the concrete signed construction approach the general integer-variable bounds?** The general representation theorem in Sun's September 2026 preprint uses ten integer unknowns for arbitrary recursively enumerable sets [11]. A complete fixed Con(ZFC) specialization should materialize or finitely specify every index and coefficient, state the effective compiler used, and compare its actual degree and description size.

6. **Which parts should be formalized first?** The rational auxiliary-Pell lemma, the two square-comparison identities, the noncircular bootstrap, and the explicit witness shifts are small reusable targets. Proving a generic finite norm evaluator correct would then connect the emitted DAG to these lemmas. The existing compiler still needs its own end-to-end formal bridge.

7. **How should minimality be stated?** A meaningful lower bound must specify domain, total degree, allowed coefficient encoding, and the arithmetic model. The current collection supports a frontier of tradeoffs, not a mathematically justified global "simplest equation."

References

[1] V. Reshetnikov, ProveIt, *Con(ZFC) in Diophantine form*, catalogue at revision `ae3007ca4467011e0e051c84acb4a1dd13971b0e`. https://github.com/VladimirReshetnikov/ProveIt/tree/ae3007ca4467011e0e051c84acb4a1dd13971b0e/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms. Local copy: `references/COMPARISON_AT_PIN.md`.

[2] ProveIt, `KEndpoint215.lean`, same revision, `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/`. Local endpoint source: `references/KEndpoint215.lean`. Literal machine SHA-256:

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

[3] ProveIt wave-3 package 19, *Twelve positive variables and materialized coefficients for the consistency of ZFC*. In particular `WANG943_COEFFICIENTS.md` and `WANG943_COMPACT_LAYOUT.md`, included under `coefficients/`.

[4] ProveIt wave-3 package 21, *A twelve-variable equation for Con(ZFC)*. In particular `NORM12_PROOF.md` and `NORM12_LOWER_DEGREE.md`, included under `references/`.

[5] ProveIt, *Independent proof and source audit: the 67-operation blank-input polynomial*. Included as `references/FIXED_BLANK_67_PROOF.md`, together with the ordinary and signed-domain interface proofs.

[6] ProveIt wave-3 package 20, *Twelve positive unknowns and materialized coefficients for ZFC consistency*. Low-degree source polynomials and their domain reduction proof are included under `low_degree/upstream/20-root12/`.

[7] ProveIt wave-3 package 22, *Twelve-variable Diophantine forms of ZFC consistency*. Transport shift proof: `references/RETAINED59_PROOF.md`.

[8] ProveIt wave-3 package 23, *Twelve positive unknowns for the consistency of ZFC*. Odd-scale proof: `low_degree/upstream/23-separating12/proofs/ODD_SCALE_27_PROOF.md`.

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[11] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026, Theorem 1.1. https://arxiv.org/abs/2609.01594.